



Figure 2: Sanskrit in LaTeX



Figure 3: BOSS logo

processed with pdfLaTeX. You can download all these files from http://opensourceforu.eytimes.com/article_source_code/march2016/os_sanskrit.zip.

As mentioned earlier, by default, the package Devanagari will process text in Sanskrit mode. But if you need to write Hindi text using the package, you must use the command `@hindi` so that the pre-processor will assume you are working with Hindi characters.

Bharat Operating System Solutions

Germans preparing a dictionary, Americans developing fonts and, finally, a Dutchman developing the transliteration scheme for Sanskrit — my frustration was mounting! Then I came across the Bharat Operating System Solutions (BOSS). Finally, this was real work being done by our countrymen to support Sanskrit and other Indian languages. BOSS is an Indian GNU/Linux distribution developed by C-DAC and is customised to suit the Indian needs. Figure 3 shows the logo of BOSS.

There are server, desktop and educational versions of BOSS available. The desktop version of BOSS is called Boss Desktop and the educational version is called EduBOSS. The latest release of the BOSS Desktop and EduBOSS is BOSS version 6, named Anoop, which uses the GNOME Desktop Environment 3.14. BOSS Desktop has many provisions for users of Indian languages. Currently, 18 Indian languages are supported by BOSS Desktop including Sanskrit, Hindi, Tamil, Telugu and Malayalam.

In BOSS, inputs can be given in Indian languages also. An input method called *Ibus* provides onscreen display for input method switching involving character sets of different Indian languages. The Orca screen reader is enabled to read out the text on the screen. But when Indian languages are read by Orca, the accent is slightly anglicised. Let us hope future versions of BOSS will give better performance.

The educational version, EduBOSS, is customised for Indian schools. There are many games available with EduBOSS. These games can be further modified to help children learn Sanskrit. For example, the game called Gbrainy offers a variety of mathematical problems. This game can be modified to include Vedic mathematical problems. Similarly, the KDE based game Kanagram, which offers puzzles based on English anagrams, can be modified to include Sanskrit anagrams. I hope somebody will take up this challenge and modify these games, so that a lot of children can have the best of both worlds - that of Sanskrit and open source software.

Indian Heritage Group of C-DAC

While searching for the details of BOSS, I came across the Indian Heritage Group (IHG) of C-DAC, whose group coordinator is Dr P. Ramanujan. IHG is responsible for Sanskrit and Vedic processing using computers. I don't remember the book in which I read that Srinivasa Ramanujan had single-handedly derived every mathematical equation ever conceived by the western intellects collectively. When I came across the works of Dr P. Ramanujan, I had the same feeling. Whatever the entire western community has done to preserve and propagate Sanskrit has been done single-handedly by Dr P. Ramanujan.

I am not qualified enough to comment on his works. My knowledge of Sanskrit is not adequate to assess his brilliant work. But I feel I am privileged to discuss the work of Dr Ramanujan in the field of Sanskrit related software. He has created a database of over 30,000 sutras from the sastras all by himself, named *Sakala Sastra Sutra Kosa*. He was also instrumental in the development of ISCII (Indian Script Code for Information Interchange), for which he developed the Vedic part. He developed a software called DESIKA to generate and analyse Sanskrit words. DESIKA software can be used for noun generation, verb generation for all tenses, grammatical analysis of words in a sentence, etc. He was also responsible for digitising a lot of old Sanskrit manuscripts. IHG has also developed other software like C-VYASA, which can be used as a Vedic editor and Sanskrit authoring system. The group has developed a searchable knowledge base of Vedas. Currently, IHG is developing software called *Swadhyaya*, which is a self-teaching system for Sanskrit.

Both western as well as Indian computer scientists have contributed to the development of computing for Sanskrit. Providing more and more computing tools for Sanskrit processing will definitely attract a large number of young people to Sanskrit. But it is not just the case of Sanskrit benefiting from computer science alone; it could be the other way round also. Paninian grammar of Sanskrit is considered as unambiguous grammar by many linguists and grammarians. In the field of Natural Language Processing (NLP), it is absolutely essential to have unambiguous grammar. Paninian grammar might give a better insight to solving this long standing problem. Similarly, the hidden treasures of Vedic mathematics could also help computer science advance further. So let us hope this mutually beneficial interaction between computer science and Sanskrit will continue and prosper in the future also. 

By: Deepu Benson

The author, who has nearly 16 years of programming experience, is currently working as assistant professor in Amal Jyothi College of Engineering, Kerala. The open source tools of his choice include ns-2 and ns-3. The author maintains a technical blog at www.computingforbeginners.blogspot.in. He can be reached at deepump@hotmail.com.